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CHM 419 — Past Questions & Materials
by Renance

Course: CHM 419
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 400

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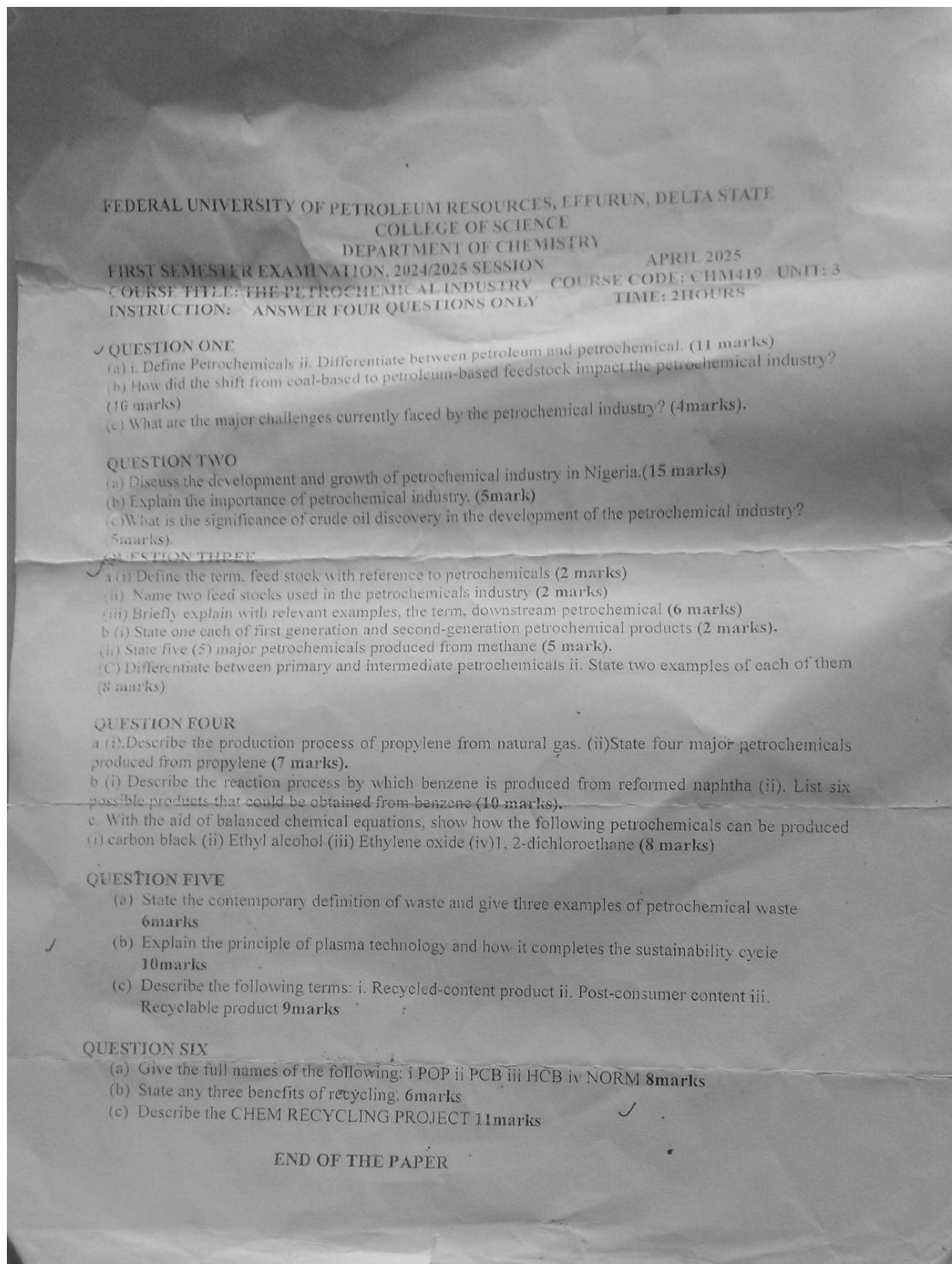
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Paper 1 of 4 — CHM 419 - 2024/2025 - First Semester - THE PETROCHEMICAL INDUSTRY

Level 200 • First Semester • PastQuestion

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Paper 2 of 4 — CHM 419 - 2024/2025 - First Semester - THE PETROCHEMICAL INDUSTRY

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
 COLLEGE OF SCIENCE
 DEPARTMENT OF CHEMISTRY
 FIRST SEMESTER EXAMINATION, 2024/2025 SESSION APRIL 2025
 COURSE TITLE: THE PETROCHEMICAL INDUSTRY COURSE CODE: CHM419 UNIT: 3
 INSTRUCTION: ANSWER FOUR QUESTIONS ONLY TIME: 2HOURS

✓ QUESTION ONE

- (a) i. Define Petrochemicals ii. Differentiate between petroleum and petrochemical. (11 marks)
 (b) How did the shift from coal-based to petroleum-based feedstock impact the petrochemical industry? (10 marks)
 (c) What are the major challenges currently faced by the petrochemical industry? (4marks).

QUESTION TWO

- (a) Discuss the development and growth of petrochemical industry in Nigeria. (15 marks)
 (b) Explain the importance of petrochemical industry. (5mark)
 (c) What is the significance of crude oil discovery in the development of the petrochemical industry? (5marks).

QUESTION THREE

- a (i) Define the term, feed stock with reference to petrochemicals (2 marks)
 (ii) Name two feed stocks used in the petrochemicals industry (2 marks)
 (iii) Briefly explain with relevant examples, the term, downstream petrochemical (6 marks)
 b (i) State one each of first generation and second-generation petrochemical products (2 marks).
 (ii) State five (5) major petrochemicals produced from methane (5 mark).
 (C) Differentiate between primary and intermediate petrochemicals ii. State two examples of each of them (8 marks)

QUESTION FOUR

- a (i). Describe the production process of propylene from natural gas. (ii) State four major petrochemicals produced from propylene (7 marks).
 b (i) Describe the reaction process by which benzene is produced from reformed naphtha (ii). List six possible products that could be obtained from benzene (10 marks).
 c. With the aid of balanced chemical equations, show how the following petrochemicals can be produced (i) carbon black (ii) Ethyl alcohol (iii) Ethylene oxide (iv) 1, 2-dichloroethane (8 marks)

QUESTION FIVE

- (a) State the contemporary definition of waste and give three examples of petrochemical waste 6marks
 (b) Explain the principle of plasma technology and how it completes the sustainability cycle 10marks
 (c) Describe the following terms: i. Recycled-content product ii. Post-consumer content iii. Recyclable product 9marks

QUESTION SIX

- (a) Give the full names of the following: i POP ii PCB iii HCB iv NORM 8marks
 (b) State any three benefits of recycling. 6marks
 (c) Describe the CHEM RECYCLING PROJECT 11marks

END OF THE PAPER

Paper 3 of 4 — CHM 419 The petrochemical Industry 2024/2025

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

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END OF THE PAPER

Paper 4 of 4 — CHM419 Petrochemical Industry - Waste

Level 400 • First Semester • Handout

1

THE PETROCHEMICAL INDUSTRY (CHM419)

History of Petrochemicals

In 1835, Henri Victor Regnault, a French chemist left vinyl chloride in the sun and found white solid at the bottom of the flask which was polyvinyl chloride. In 1839, Eduard Simon discovered polystyrene by accident by distilling storax. In 1856, William Henry Perkin discovered the first synthetic dye, Mauveine. In 1888, Friedrich Reinitzer, an Austrian plant scientist observed cholesteryl benzoate had two different melting points. In 1909, Leo Hendrik Baekeland invented bakelite made from phenol and formaldehyde. In 1920, Union Carbide built in West Virginia first petrochemical plant in the world. In 1928, synthetic fuels were invented using Fischer-Tropsch process. In 1929, Walter Bock invented synthetic rubber Buna-S which is made up of styrene and butadiene and used to make car tires. In 1933, Otto Röhm polymerized the first acrylic glass methyl methacrylate. In 1935, Michael Perrin invented polyethylene. In 1937, Wallace Hume Carothers invented nylon. In 1938, Otto Bayer invented polyurethane. In 1941, Roy Plunkett invented Teflon. In 1946, he invented Polyester. Polyethylene terephthalate (PET) bottles are made from ethylene and paraxylene. In 1949, Fritz Stastny turned polystyrene into foam. After World War II, polypropylene was discovered in the early 1950s. In 1965, Stephanie Kwolek invented Kevlar.

The petrochemical industries produce huge quantities of oily sludge during the crude oil refining process that emanates from different sources such as slop oil emulsions, crude oil tank bottoms, onsite effluent treatment ponds, and oil or water separator.

The widespread use of petrochemical produce has contaminated all spheres of life and has deteriorated all environmental resources especially the petrochemical production zone leading to the contamination of soil, water and air. The wastewater effluent of typical petrochemical refinery comprised of diverse organic compounds in very high concentration and is less degradable in the environment

The industry equally produces a large amount of solid oily waste, which is hazardous and can't be recycled also. Primarily as PAH, BTEX (benzene, toluene, ethylbenzene, and xylene) compounds produced from the petroleum industry have high water solubility, thus contaminating the first water (fine water or clear water). Other hydrocarbon contaminants are hydrophobic and have low solubility in water. Thus, their uptake by microorganisms through natural processes is slow.

2

What is waste?

What is anything or material that would otherwise be thrown away as trash

Hazardous Waste

Any industrial by-product, especially from the manufacture of chemicals, that is destructive to

the environment or dangerous to the health of people or animals.

Understanding the Four Characteristics of Hazardous Waste

When it comes to hazardous waste disposal and management, understanding the waste your organization generates is imperative. Hazardous waste is heavily regulated, and thus cannot just be tossed out with your everyday trash. To know if your organization is handling hazardous waste, the first step is to assess its characteristics.

When categorizing hazardous waste, the Environmental Protection Agency (EPA) breaks it down by four characteristics:

- ignitability, or something flammable
- corrosivity, or something that can rust or decompose
- reactivity, or something explosive
- toxicity, or something poisonous

These high level categories each have their own characteristics that further help you as a generator define what you are dealing with.

Ignitability

There are three types of ignitable forms:

- Liquids with a flash point—the lowest temperature at which fumes above waste ignite—of 60 degrees Celsius or 140 degrees Fahrenheit. Examples include alcohol, gasoline, and acetone.
- Solids that spontaneously combust.
- Oxidizers and compressed gasses.

Corrosivity

Corrosive substances, such as hydrochloric acid, nitric acid, and sulfuric acid, have the ability eat through containers, causing the leakage of harmful materials. A corrosive is anything liquid with a pH of less than or equal to 2 or greater than or equal to 12.5, or has the ability to corrode steel. Everyday example of corrosives include battery acid and rust removers.

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Reactivity

Given their instability, reactive wastes can be very dangerous. The EPA recognizes that there are too many conditions and situations to identify all types of reactive materials. However, they use the following as guidelines to assist generators:

- unstable, and routinely experiences violent change without detonating
- potential for explosive mixture or violent reaction when combined with water
- toxic gases are released when mixed with water

Toxicity

Poisonous materials pose a threat to our groundwater, which can have long term effects to human health and the environment. This is different from the first three characteristic groups, which the EPA views as containing immediate and firsthand dangers. There are 60 contaminants on the

toxicity characteristics list. These contaminants are identified solely through a test method called Toxicity Characteristic Leaching Procedure or TCLP.

As a generator of any the above, you have two options to determine which characteristics above best define the waste you generate: test it, or use applied knowledge from previous company records or industry data and studies.

Examples of Petrochemical Waste

- Oily Sludges generated during oil tank cleaning operations
- Hazardous Waste Streams Contaminated with Heavy Metals and Persistent Organic Pollutants (POPs; e.g. PCBs and HCBs)
- Secondary residues from alternative treatment processes
- Naturally Occurring Radioactive Materials (NORM)
- Acid tars generated from benzole refining, oil re-refining and white oil production

Plasma Technology in the Treatment of Waste

The heat and ultra-violet light properties of plasma technology treats contaminated petrochemical waste materials so that the pollutant is destroyed and the bulk mass material is recovered as a safe, inert building product, so completing the sustainability cycle.

This attitude marks something of a reversal for the petrochemicals industry, which supplies the base chemicals and resins needed to manufacture a wide range of plastic products from bags and bottles to car parts and construction materials. Producers for years took a hands-off approach to waste management, arguing that consumers bore the greatest responsibility in ensuring proper disposal of the plastics they used.

4

WASTE RECYCLING

Recycling is the process of collecting and processing materials that would otherwise be thrown away as trash and turning them into new products. Recycling can benefit communities and the environment

STEPS IN RECYCLING

1. Collection and Processing

There are several methods for collecting recyclables, including curbside collection, drop-off centers, and deposit or refund programs.

After collection, recyclables are sent to a recovery facility to be sorted, cleaned and processed into materials that can be used in manufacturing. Recyclables are bought and sold just like raw materials would be, and prices go up and down depending on supply and demand in the United States and the world.

2. Manufacturing

More and more of today's products are being manufactured with recycled content. Common household items that contain recycled materials include the following:

- Newspapers and paper towels
- Aluminum, plastic, and glass soft drink containers
- Steel cans
- Plastic laundry detergent bottles

Recycled materials are also used in new ways such as recovered glass in asphalt to pave roads or recovered plastic in carpeting and park benches.

3. Purchasing New Products made from Recycled Materials

You help close the recycling loop by buying new products made from recycled materials. There are thousands of products that contain recycled content. When you go shopping, look for the following:

- Products that can be easily recycled
- Products that contain recycled content

Below are some of the terms used in recycling

- Recycled-content product - The product was manufactured with recycled materials either collected from a recycling program or from waste recovered during the normal manufacturing process. The label will sometimes include how much of the content was from recycled materials.

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- Post-consumer content - Very similar to recycled content, but the material comes only from recyclables collected from consumers or businesses through a recycling program.
- Recyclable product - Products that can be collected, processed and manufactured into new products after they have been used. These products do not necessarily contain recycled materials. Remember not all kinds of recyclables may be collected in your community so be sure to check with your local recycling program before you buy.

Some of the common products you can find that can be made with recycled content include the following:

- Aluminum cans
- Car bumpers
- Carpeting
- Cereal boxes
- Comic books
- Egg cartons
- Glass containers
- Laundry detergent bottles
- Motor oil
- Nails
- Newspapers
- Paper towels
- Steel products

- Trash bags

BENEFITS OF RECYCLING

- Reduces the amount of waste sent to landfills and incinerators
- Conserves natural resources such as timber, water and minerals
- Increases economic security by tapping a domestic source of materials
- Prevents pollution by reducing the need to collect new raw materials
- Saves energy
- Supports manufacturing and conserves valuable resources
- Helps create jobs in the recycling and manufacturing industries

CHEMICAL RECYCLING OF PLASTIC WASTE

From plastic waste to new chemical products

Plastics do have proven benefits during their use phase – for example preservation of food loss, lightweight construction of vehicles and building insulation. Plastic waste, however, and in particular plastic waste in the context of marine littering, is perceived as a major global challenge. There is also increasing regulatory pressure regarding recycling quota and recyclability on the one hand and strong commitments of our customers towards increasing the share of recycled material in their offerings on the other hand. Solving these challenges requires innovation and joint efforts globally across the value chain. We will contribute to tackling this.

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Since mechanical recycling is limited because of an increase of residues in the material in each cycle, a team across BASF has taken up this challenge and developed the Chem Cycling project. With chemical recycling, fossil resources for chemical production can be replaced with recycled material from plastic waste. How does it work? Plastic waste will be transformed into a raw material using thermochemical processes. The raw material can be fed into the Verbund to create new chemical products with excellent product performance based on recycled plastic waste.

Chemical recycling can help reduce the proportion of plastic waste which ends up in landfill or incineration. However, thermochemical recycling needs acceptance as recycling from market and regulators. There are many open questions with regards to technology, economy and regulation. We closely engage with the relevant stakeholders to tackle these challenges.

With eco-efficiency analysis we ensure that the innovative approach creates value for the environment. Moreover, chemical recycling represents an exciting business opportunity for us and our customers, as the resulting products are of equal quality to the products derived from fossil feedstock. We are currently working with customers to produce the first prototypes for customer products with chemically recycled material.